EECE 5639: Computer Vision
Fall 2026, Homework 2

Assigned: Wed, Sept 30, 2026 **Points: 100** **Due: Fri, Oct 09, 2026**

1. **[30 points]** We have seen in class that certain 2-D smoothing filters such as the Box filter and the Gaussian filter are *separable*. They can be implemented by applying a 1-D filter along the rows followed by a 1-D filter along the columns. In this problem we will consider a 1-D, noisy image and find out which linear filter minimizes the output noise variance.

   Consider three consecutive pixels of a 1-D image with true gray values $\hat{I}_{i-1}, \hat{I}_i, \hat{I}_{i+1}$ respectively. The observed image is corrupted by additive white Gaussian noise; that is, $I_i = \hat{I}_i + N_i$, where $N_i \sim \mathcal{N}(0, \sigma^2)$. The noise samples at different pixels are independent. The noisy image is smoothed using a 3-pixel filter $\begin{bmatrix} w_1 & w_2 & w_3 \end{bmatrix}$. The filtered pixel value is

   $$I_i^{\text{out}} = w_1 I_{i-1} + w_2 I_i + w_3 I_{i+1}.$$

   To preserve the average image intensity, assume that the filter weights satisfy $w_1 + w_2 + w_3 = 1$.

   (a) **[5 points]** Express the noise component of I_i^{out} (i.e., the output noise) in terms of N_{i-1}, N_i, N_{i+1} and the filter weights w_1, w_2, w_3.

   (b) **[5 points]** Determine the variance of the output noise as a function of w_1, w_2, w_3, σ.

   (c) **[10 points]** Find the filter weights w_1^*, w_2^*, w_3^* that minimize the output noise variance subject to $w_1 + w_2 + w_3 = 1$.

   NOTE: This is a "constrained optimization" problem, and the traditional approach to solve such problems is to use Lagrange multipliers. Suppose that we want to minimize $f(x_1, x_2, ..., x_n)$ subject to the constraint $g(x_1, x_2, ..., x_n) = 0$. The first step is to define the Lagrangian

   $$L(x_1, x_2, ..., x_n; \lambda) = f(x_1, x_2, ..., x_n) + \lambda g(x_1, x_2, ..., x_n).$$

   The parameter λ is called the Lagrange multiplier. We find candidate optimal points by equating all partial derivatives of the Lagrangian to zero. That is,

   $$\frac{\partial L}{\partial x_1} = 0; \quad \frac{\partial L}{\partial x_2} = 0; ...; \frac{\partial L}{\partial x_n} = 0; \frac{\partial L}{\partial \lambda} = 0.$$

   We solve the resulting equations for $x_1, x_2, ..., x_n$ and λ. Follow this method for the image smoothing problem.

   (d) **[2 points]** What familiar smoothing filter do these optimal weights correspond to?

   (e) **[3 points]** Using this smoothing filter, by what factor has the noise variance been reduced compared with the original noisy image?

(f) **[5 points]** The optimal filter obtained in part (c) minimizes the output noise variance. However, noise reduction is not the only consideration when smoothing an image. What undesirable effect might this filter have on image features such as edges and fine details? Why might a Gaussian filter sometimes be preferred?

2. **[40 points]** In this problem, you will compare in simulation three commonly used image smoothing filters: the **Box filter**, the **Gaussian filter**, and the **Median filter**. You may use existing MATLAB or Python functions to implement these filters. You are **not** required to implement the filtering algorithms from scratch. The objective is to investigate how the three filters affect an image and how their performance depends on the type of noise present.

   The deliverables are - (i) a screenshot of your MATLAB or Python code attached to your homework, (ii) figures as requested below, (iii) observations/explanations as requested below.

   Use the grayscale image `cameraman.tif` for all parts of this problem so that you can directly compare the effects of the different filters and noise types.

   (a) **[6 points]** Normalize the image so that its pixel intensities lie in the range $[0, 1]$. Add independent zero-mean Gaussian noise to each pixel with standard deviation $\sigma = 0.05$:

   $$I(x, y) = \hat{I}(x, y) + N(x, y), \qquad N(x, y) \sim \mathcal{N}(0, 0.05^2).$$

   Apply the following filters to the noisy image:

   - a 5×5 Box filter,
   - a 5×5 Gaussian filter with $\sigma = 1$, and
   - a 5×5 Median filter.

   Display the noisy image and the three filtered images side-by-side.

   (b) **[2+2+2 = 6 points]** Based on your observations from part (a), answer the following questions:

   i. Which filter appears to remove the Gaussian noise most effectively?

   ii. Does the filter that removes the most noise also produce the best-looking image? Base your answer on what happens to edges and other fine details.

   iii. The Box and Gaussian filters both compute weighted averages of neighboring pixels. Explain why they produce different results.

   (c) **[6 points]** Starting again from the original image, normalize the image so that its pixel intensities lie in the range $[0, 1]$. Add salt-and-pepper noise with a noise density of $d = 0.05$. Thus, approximately 5% of the image pixels should be corrupted by salt-and-pepper noise. Apply the same three filters to the noisy image. Display the noisy image and the three filtered results side-by-side.

   (d) **[2+2+2 = 6 points]** Based on your observations from part (c), answer the following questions:

   i. Which filter appears to remove the salt-and-pepper noise most effectively?

   ii. Explain why this filter is effective at removing salt-and-pepper noise.

   iii. How does the performance of this filter in the current experiment compare with its performance in the Gaussian-noise experiment in parts (a)–(b)?

(e) **[6 points]** Revisit the image containing Gaussian noise. Apply a Gaussian filter using a fixed 7×7 kernel and the following values of σ:

$$\sigma = 0.6, \qquad \sigma = 1, \qquad \sigma = 2.$$

Display the noisy image and the three filtered images side-by-side.

(f) **[5 points]** Based on the results from part (e), discuss how increasing σ affects

- noise reduction,
- preservation of fine image details,
- sharpness of edges.

Explain whether the trend is consistent with the weights of the Gaussian filter for varying σ.

(g) **[5 points]** Based on parts (a) – (f) above, list 2-3 major conclusions on how the appropriate choice of filter depends on - (i) the type of noise, and (ii) the desired balance between noise reduction and preservation of image details.

3. **[30 points]** In this problem, you will compare three commonly used edge detection methods: the **Prewitt**, **Sobel**, and **Canny** edge detectors. You may use existing MATLAB or Python functions to implement the edge detectors. You are **not** required to implement the algorithms from scratch. The objective is to investigate how the three methods identify edges in an image and how their performance is affected by image noise. The deliverables are - (i) a screenshot of your MATLAB or Python code attached to your homework, (ii) figures as requested below, (iii) observations/explanations as requested below.

(a) **[6 points]** Apply the Sobel edge detector to the original image. Compute the image gradients in both the x- and y-directions, denoted by G_x and G_y, respectively. Then combine the two gradient components to obtain the gradient magnitude $G = \sqrt{G_x^2 + G_y^2}$.

Display the original image and the gradient images $|G_x|$, $|G_y|$, and G side-by-side. Scale the gradient images appropriately for visualization. Briefly comment on what types of edges are emphasized by $|G_x|$ and $|G_y|$, and how the gradient magnitude G combines the information from both directions.

(b) **[6 points]** Now apply the Prewitt and the Canny edge detectors to the original image. For the Prewitt, compute the gradient magnitude $G = \sqrt{G_x^2 + G_y^2}$. For the Canny edge detector, use the default threshold settings provided by your MATLAB or Python implementation. Display the original image, the Prewitt gradient magnitude image, the Sobel gradient magnitude image, and the Canny edge image side-by-side.

(c) **[6 points]** Compare the Prewitt, Sobel, and Canny results from part (b). Keep in mind that the Prewitt and Sobel results shown here are gradient-magnitude images, whereas Canny performs additional processing to produce a final edge map. In particular, comment on

- how clearly the major object boundaries are detected,
- the thickness and continuity of the detected edges, and
- any noticeable differences among the three methods.

(d) **[6 points]** Normalize the image and add Gaussian noise according to:

$$I(x, y) = \hat{I}(x, y) + N(x, y)$$

with $N(x, y) \sim \mathcal{N}(0, 0.05^2)$. Apply the Prewitt, Sobel, and Canny edge detectors to the noisy image in the same manner as in part (b). Use the same Canny settings for the original and noisy images. Display the noisy image, the Prewitt gradient magnitude, the Sobel gradient magnitude, and the Canny edge image side-by-side.

(e) **[1+1+4 = 6 points]** Compare the results from the noisy image in part (d) with those obtained from the original image in part (b). In particular, answer the following questions:

i. Which method appears to be the most sensitive to the added noise?

ii. Which method appears to be the most robust to the added noise?

iii. Explain why the three edge detectors respond differently to Gaussian noise based on how each method operates. In particular, consider the amount of smoothing performed before or during the computation of image gradients.